Introduction to LLMs: Take Home Assignment 1

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Other Submitted Files:

- Python file of the source code: q2_retrieval_compact.py
- Dataset file: data/documents.csv (300 Documents with column names being "ID " and "Text")

GitHub repository: https://github.com/coolprash06/IntroToLLMs_THA1_Question2

1.Results

1.1 Introduction

Systems A, B and C were tested on the same 10 queries. For each of queries, the three systems returned its top 3 documents based on their ranking (They are ranked by the similarity score). The list of relevant documents for their respective queries has been found by manually reading the headlines before running any of the 3 systems. Hence a returned document counts as correct only when it is part of the set of relevant documents for that query.

1.2 - Top-3 results by query type:

1.2.1 - Exact Keyword Queries

```
Q1 | Exact keyword | oil prices
   System A: 203* (0.0952), 219* (0.0952), 170* (0.0870)
   System B: 219* (0.0652), 170* (0.0612), 151* (0.0500)
   System C: 151* (0.6340), 170* (0.5889), 159* (0.5417)

Q2 | Exact keyword | Firefox
   System A: 252* (0.0588), 279* (0.0526), 237* (0.0417)
   System B: 252* (0.0286), 279* (0.0256), 237* (0.0208)
   System C: 237* (0.5699), 279* (0.5273), 252* (0.5251)
```

Figure 1 (Section 2.2.1).

1.2.2- Paraphrased Queries

```
Q3 | Paraphrase | struggling airlines trying to avoid bankruptcy
   System A: 167* (0.0909), 224* (0.0870), 201* (0.0741)
   System B: 167* (0.0638), 224* (0.0625), 201* (0.0345)
   System C: 180* (0.6222), 165* (0.4823), 162* (0.4750)

Q4 | Paraphrase | privately built plane flies to the edge of space
   System A: 285* (0.0690), 298* (0.0588), 242 (0.0476)
   System B: 285* (0.0339), 298* (0.0278), 242 (0.0222)
   System C: 285* (0.5240), 298* (0.4428), 217 (0.3747)
```

1.2.3- Queries Using Synonyms:

```
Q5 | Synonym | joblessness rate falls
   System A: 163* (0.0345), 261 (0.0345), 212 (0.0333)
   System B: 212 (0.0169), 261 (0.0167), 163* (0.0161)
   System C: 177* (0.5286), 163* (0.5222), 233 (0.4154)

Q6 | Synonym | computer virus attacks
   System A: 209 (0.0500), 287* (0.0435), 18 (0.0417)
   System B: 209 (0.0250), 47 (0.0208), 184 (0.0204)
   System C: 225* (0.4524), 230* (0.3738), 287* (0.3708)
```

1.2.4- Word Order Sensitive Queries:

```
Q7 | Word order | Braves beat Astros
   System A: 139* (0.0952), 137* (0.0938), 50 (0.0625)
   System B: 137* (0.0746), 139* (0.0444), 50 (0.0303)
   System C: 137* (0.8249), 139* (0.6760), 118 (0.5172)

Q8 | Word order | Astros beat Braves
   System A: 139 (0.0952), 137 (0.0938), 50 (0.0625)
   System B: 139 (0.0444), 137 (0.0435), 50 (0.0303)
   System C: 137 (0.8083), 139 (0.6521), 118 (0.5162)
```

1.2.5- Ambiguous or Difficult queries

```
Q9 | Ambiguous | Sun
   System A: 275 (0.0400), 227* (0.0357), 240* (0.0357)
   System B: 275 (0.0196), 227* (0.0182), 240* (0.0172)
   System C: 227* (0.2940), 240* (0.2556), 231 (0.2456)

Q10 | Ambiguous | hurricane
   System A: 95* (0.0588), 67* (0.0385), 158* (0.0357)
   System B: 95* (0.0256), 67* (0.0196), 158* (0.0172)
   System C: 52* (0.5867), 95* (0.4253), 67* (0.4117)
```

1.3 - N-gram experiment

```
N-gram experiment (System B)
   [1]        relevant in top 3: 20 / 30   queries with no match: 0
   [2]        relevant in top 3: 6 / 30   queries with no match: 7
   [3]        relevant in top 3: 1 / 30   queries with no match: 9
   [1, 2]     relevant in top 3: 19 / 30   queries with no match: 0
   [1, 2, 3]  relevant in top 3: 19 / 30   queries with no match: 0
```

1.4 - Final comparison

id	type	query	System A	System B	System C
Q1	Exact keyword	oil prices	3 / 3	3 / 3	3 / 3
Q2	Exact keyword	Firefox	3 / 3	3 / 3	3 / 3
Q3	Paraphrase	struggling airlines trying to avoid bankruptcy	3 / 3	3 / 3	3 / 3
Q4	Paraphrase	privately built plane flies to the edge of space	2 / 3	2 / 3	2 / 3
Q5	Synonym	joblessness rate falls	1 / 3	1 / 3	2 / 3
Q6	Synonym	computer virus attacks	1 / 3	0 / 3	3 / 3
Q7	Word order	Braves beat Astros	2 / 3	2 / 3	2 / 3
Q8	Word order	Astros beat Braves	0 / 3	0 / 3	0 / 3
Q9	Ambiguous	Sun	2 / 3	2 / 3	2 / 3
Q10	Ambiguous	hurricane	3 / 3	3 / 3	3 / 3
TOTAL			20 / 30	19 / 30	23 / 30

2. Discussion

2.1 Method

2.1.1 Dataset

The dataset has 300 short English News Documents taken from the public **AG News** dataset which is available on Hugging face. Each document has 2 parts: an id and a short text. The id number ranges from 0 to 299 and it identifies the document. This is used to refer to the document at the time of result and in the lists of relevant documents. The text is a headline followed by a short summary. Both the headline and the short summary is stored in one text field. The test queries were written about the topics which appeared in the documents.

An Example Document:

ID	headline
151	*Oil Leaps as U.S. Inventories Dwindle* LONDON (Reuters) - Oil prices jumped a dollar on Thursday after the U.S. government said crude stocks fell to a fresh five-month low last week and distillate fuel inventories barely grew ahead of winter.

2.1.2 Test queries

10 Test queries were tested and all of these queries were written for text which appear in the dataset. The 10 test queries were divided into 2 for each of the following:

- Exact keyword queries
- Paraphrased queries
- Queries using synonyms
- Word order sensitive queries
- Ambiguous or difficult queries

```
# --- QUERIES: start ---
QUERIES = [
    ("Q1", "Exact keyword", "oil prices", [73, 151, 159, 170, 203, 219]),
    ("Q2", "Exact keyword", "Firefox", [237, 252, 279]),
    ("Q3", "Paraphrase", "struggling airlines trying to avoid bankruptcy", [162, 164, 165, 167, 180, 201, 224]),
    ("Q4", "Paraphrase", "privately built plane flies to the edge of space", [238, 265, 285, 298]),
    ("Q5", "Synonym", "joblessness rate falls", [158, 163, 177]),
    ("Q6", "Synonym", "computer virus attacks", [225, 230, 246, 283, 287]),
    ("Q7", "Word order", "Braves beat Astros", [137, 139]),
    ("Q8", "Word order", "Astros beat Braves", []),
    ("Q9", "Ambiguous", "Sun", [199, 227, 240]),
    ("Q10", "Ambiguous", "hurricane", [52, 67, 95, 158]),
]
# --- QUERIES: end ---
```

The figure shows the 10 test queries and their relevant documents.

For the query:

("Q1", "Exact keyword", "oil prices", [73, 151, 159, 170, 203, 219])

Part	Meaning
"Q1"	The query's id
"Exact keyword"	Which of the five query types it tests
"oil prices"	The query text the systems search with
[73, 151, 159, 170, 203, 219]	The ids of the relevant documents

The relevant documents were decided manually before running any of the systems. A document only counts if the query's subject plays a key role in the short text. The systems only receive the query text. The list containing the ids of relevant documents is used as an answer key to count how many of the top 3 results are relevant.

2.1.3- Systems A, B & C:

All 3 systems receive the same query and score all 300 documents and return the top 3 highest scoring documents.

System A: Keyword Retrieval

The text is converted to lowercase and anything that is not a letter or number is treated as a space thus splitting the text into words. All stopwords (ex: "the", "and", "of"…) are removed. The remaining words form a set with each word once and without any order. Each document is scored with a Jaccard Similarity,

$$\text{Jaccard}(Q, D) = \frac{|Q \cap D|}{|Q \cup D|} = \frac{\text{words in both}}{\text{words in either}}$$

Here, Q= Set made from the Query

D= Set made from the document

The following is the equivalent python method:

```
48  def jaccard(set1, set2):
49      union = set1 | set2
50      if len(union) == 0:
51          return 0
52      return len(set1 & set2) / len(union)
```

System B: N Gram retrieval:

System B follows the same preparation as System A but the set contains unigrams and bigrams. Unigrams are single words and bigrams are a pair of neighbouring words.

Example: For query 7: “Braves beat Astros”, The set would become:

{braves, beat, astros, *braves beat*, *beat astros*}

Choice of n-grams: Five n-gram settings were tested (Shown in section 1.3). Bigrams alone found no match for 7 out of the 10 queries and trigrams alone did not find results for 9 of the 10 queries. Also adding trigrams to unigrams and bigrams made no difference (19/30 in both cases). Hence unigrams and bigrams were chosen, as they matched every query while adding sensitivity to word order.

System B uses the same Jaccard formula as System A but the set made from the Query would be different for both the systems.

System C - Embedding Retrieval:

The pretrained model used in System C is “all-MiniLM-L6-v2” . It is a sentence transformer. It turns each document and the query into a 384 dimensional vector. No training is done on the model. Each document is scored with a cosine similarity.

```
def cosine_similarity(a, b):
    dot = 0.0
    sum_a = 0.0
    sum_b = 0.0
    for i in range(len(a)):
        dot = dot + a[i] * b[i]
        sum_a = sum_a + a[i] * a[i]
        sum_b = sum_b + b[i] * b[i]
    return dot / (math.sqrt(sum_a) * math.sqrt(sum_b))
```

Ranking for all 3 Systems:

Documents are sorted by score, highest first, and the top 3 are returned

Tie Breaker Case: If two documents have exactly the same score, the one with the **lower id** comes first

2.2 Comparison of the top-3 results

2.2.1 Exact keyword queries (Q1, Q2)

```
Q1 | Exact keyword | oil prices
   System A: 203* (0.0952), 219* (0.0952), 170* (0.0870)
   System B: 219* (0.0652), 170* (0.0612), 151* (0.0500)
   System C: 151* (0.6340), 170* (0.5889), 159* (0.5417)

Q2 | Exact keyword | Firefox
   System A: 252* (0.0588), 279* (0.0526), 237* (0.0417)
   System B: 252* (0.0286), 279* (0.0256), 237* (0.0208)
   System C: 237* (0.5699), 279* (0.5273), 252* (0.5251)
```

Query 1: "oil prices"

Relevant Documents: 73, 151, 159, 170, 203, 219

All the 3 Systems returned only relevant documents.

Together all 3 systems found 5 separate documents. Document 73 was however not returned by any of the systems. In System A for Query 1, Documents 203 and 219 had exactly the same score but 203 is listed first as it has lower ID number.



Query 2: "Firefox"

Relevant documents: 237, 252, 279

All three systems for Query Q2 returned top 3 Documents.

On both exact keyword queries, all three systems scored 3/3. The only differences were which relevant documents were chosen (Q1) and their order (Q1 and Q2).

2.2.2 Paraphrased queries (Q3 & Q4)

```
Q3 | Paraphrase | struggling airlines trying to avoid bankruptcy
   System A: 167* (0.0909), 224* (0.0870), 201* (0.0741)
   System B: 167* (0.0638), 224* (0.0625), 201* (0.0345)
   System C: 180* (0.6222), 165* (0.4823), 162* (0.4750)

Q4 | Paraphrase | privately built plane flies to the edge of space
   System A: 285* (0.0690), 298* (0.0588), 242 (0.0476)
   System B: 285* (0.0339), 298* (0.0278), 242 (0.0222)
   System C: 285* (0.5240), 298* (0.4428), 217 (0.3747)
```

Query 3: "struggling airlines trying to avoid bankruptcy"

Relevant Documents: 162, 164, 165, 167, 180, 201, 224

All 3 scored 3/3 but C returned an entirely different set from A and B. All 3 systems together found 6 of the 7 relevant documents. Both A & B found documents sharing the query's own words (*avoid*, *bankruptcy*).

Query 4: "privately built plane flies to the edge of space"

Relevant Documents: 238, 265, 285, 298

All the 3 systems scored only 2/3. The 3rd document returned by the systems was not relevant A and B's 3rd returned document was 242 (a space station diet), which matched only the word "space".
C's third was 217 (a commercial flight to Vietnam) which was purely about a plane flying and was not related to space.

2.2.3 Queries using synonyms (Q5 & Q6)

```
Q5 | Synonym | joblessness rate falls
   System A: 163* (0.0345), 261 (0.0345), 212 (0.0333)
   System B: 212 (0.0169), 261 (0.0167), 163* (0.0161)
   System C: 177* (0.5286), 163* (0.5222), 233 (0.4154)

Q6 | Synonym | computer virus attacks
   System A: 209 (0.0500), 287* (0.0435), 18 (0.0417)
   System B: 209 (0.0250), 47 (0.0208), 184 (0.0204)
   System C: 225* (0.4524), 230* (0.3738), 287* (0.3708)
```

Query 5: "joblessness rate falls"

Relevant Documents: 31, 158, 163, 177

Both Systems A & B returned the same documents but in a different order. Both these systems for this query scored 1/3. Document 261 is about a Music store that falls short and Document 212 interest rates
System C scored 2/3 for this query. Document 233 is about IT job growth which is related to jobs, but not about the jobless rate fall.

Query 6: "computer virus attacks"
Relevant Documents: 225, 230, 246, 279, 283, 287
System A scored 1/3. Its irrelevant results were from documents 209 (heart attacks) and 18 (car-bomb attacks).
System B scored 0/3 for this query. Its irrelevant results were from 209, 47 (arrests in Madrid), 184 (Intel shares).
System C found spyware, malware, and phishing articles although none of the documents contains the word "virus"

2.2.4 Word order sensitive queries (Q7 & Q8)

```
Q7 | Word order | Braves beat Astros
   System A: 139* (0.0952), 137* (0.0938), 50 (0.0625)
   System B: 137* (0.0746), 139* (0.0444), 50 (0.0303)
   System C: 137* (0.8249), 139* (0.6760), 118 (0.5172)

Q8 | Word order | Astros beat Braves
   System A: 139 (0.0952), 137 (0.0938), 50 (0.0625)
   System B: 139 (0.0444), 137 (0.0435), 50 (0.0303)
   System C: 137 (0.8083), 139 (0.6521), 118 (0.5162)
```

Query 7: "Braves beat Astros" (True)

Relevant Documents: 137, 139
All 3 Systems scored 2/3. All of the Systems returned documents 137 and 139.

```
137,Braves Beat Astros 6-5 to Set Up Atlanta Finale HOUSTON (Reuters) - Adam LaRoche crushed a game-tying three-run homer and J.D. Drew s
138,"Vanek, Sabres Sign Forward Thomas Vanek agreed to a three-year deal with the Buffalo Sabres on Friday, passing up his final two year
139,"Atlanta Forces Game 5 The Braves prevailed, 6-5, after coming back from a three-run deficit against the Houston Astros after Roger C
```

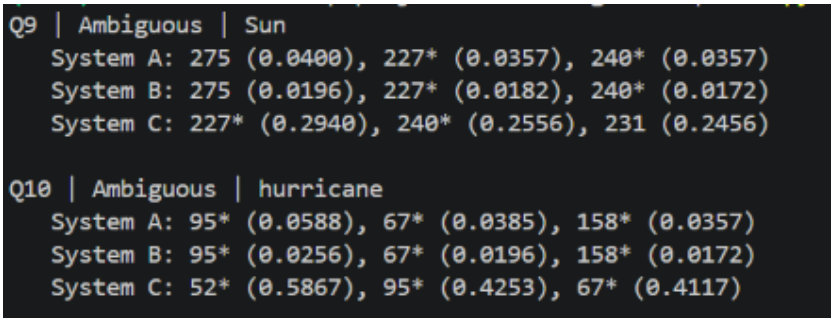
The third result for System A and System B was document 50, which was on a cricket match. System C's third result was document 118 ("Boston vs. Anaheim"), which is a different baseball game.

```
50,Cricket: England defeat India Andrew Flintoff is the hero as England beat India by 70 runs at The Oval to win the NatWest Challenge.
```

Query 8: **"Astros beat Braves" (False)**
Relevant Documents: None
All 3 Systems scored 0/3. System 'A' gave identical scores for both Query 7 and Query 8. System B scored document 137 at 0.0746 for Q7 but 0.0435 for Q8.
One of the documents returned by C was document 118 (which was about a baseball game)
For System C both the query vectors are almost identical as a result the model treats the sentences as nearly the same.

```
120   118,Boston vs. Anaheim The plot: The Anaheim Angels seemed to capture in the final week of the season the swagger they had when they won
```

2.2.5 Ambiguous or difficult queries (Q9 & Q10)



Query 9: "Sun"
Relevant Documents: 199, 227, 240
All 3 Systems scored 2/3.
A and B ranked Document 275 first. It mentions "our sun" only in passing, comparing star sizes.
C ranked the two Sun Microsystems documents first, then 231 (a space mission, mentioning the "solar system").
With the one-word query "sun", Systems A and B score each document containing "sun" as 1 divided by its number of different words, so the document with the fewest different words ranks highest. Document 275 has the fewest (25), so it came first.

Query 10: “hurricane”
Relevant Documents: 52, 67, 95, 158
All Systems scored 3/3 and found relevant documents.
All the 4 documents contain *hurricane*, so A and B again ranked them by length
System C ranked by meaning, putting the weather story itself first.

2.3 Analysis:

2.3.1 When does keyword matching work well or fail?

Keyword matching works well when the query uses the same words as in the relevant documents and the words are specific and unambiguous. Queries 1,2 and 10 scored 3/3 when tested with System A.
However, it fails when words are of same meaning but are different words. For Example, in Query 5, “joblessness” never matched “jobless” or “unemployment” and hence scored 1/3 for System A.
It also fails in the case where a word has a wrong meaning. For Query 6, the keyword “attacks” returned document 209 which is about a painkiller's heart-attack risk and document 18 which contains the text “Car-bomb attacks in Baghdad kill 17”.
Another case where it fails is when the word order matters. For Query 7 and Query 8, it returned identical results and scores.
The reasons for such failures can be attributed to the fact that System A matches exact words only. It has no knowledge of meanings, context, synonym and order.

2.3.2 What advantage do n grams provide over individual keywords?

Bigrams keep information about which words appear next to each other, which single words lose. This helps match documents with identical word order. For Query 7, document 137 was ranked first because it contains the pairs “braves beat” and “beat astros”. As a result, System B gave a score of 0.0746 to Document 137.
For Query 8, the pairs “astros beat” and “beat braves” do not appear in document 137, so its score drops to 0.0435. However, System B still returned document 137 for Query 8, as the single words (unigrams) matched. This was because System B compares both Unigrams and Bigrams.

2.3.3 Can embeddings retrieve relevant documents when different words are used?

Each text is represented as an embedding, a 384-dimensional vector. So, texts with similar meanings get vectors that point in similar directions even if the words are different.
Query 6 which contained the text, “computer virus attacks”, is a strong example to demonstrate the behaviour that embeddings retrieve relevant documents when different words are used. No document in the dataset contains the word “virus”. System C scored 3/3. It returned documents 225 ("Microsoft to offer software to help users combat spyware/malware"), 230 ("Spyware on My Machine? So What?") and 287 ("Universal browser hole a gift for scammers"). It recognised that spyware, malware, and security holes belong to the same idea as computer viruses.
For Query 5 ("joblessness rate falls"), System C ranked document 177 ("Unemployment Drops to 5.4 Percent") first. Document 177 was not in the top 3 of System A or System B.
Hence embeddings can retrieve relevant documents that use different words, which is an advantage over word matching which is used in Systems A and B.

2.3.4 Cases where the embedding system produced unexpected or incorrect results

Case 1:

Query 8: “Astros beat Braves”
Query 8 is a false statement, as document 137 states that “Braves Beat Astros 6-5”. Yet System C retrieved document 137 with a score of 0.8083. This is almost the same as document 137’s score of 0.8249 for Query 7 (“Braves beat Astros”). Both query vectors are almost identical with a cosine similarity of **0.9909**. The model captures what the sentence is about (the two teams, baseball, a win) but largely loses who did what to whom. This shows a weakness of System C compared with System B. The simpler System B separated the two queries more clearly than System C.

Case 2:

Query 9: “Sun”
System C returned document 231 at rank 3:

Document 231,"Scientists Wrap Up Genesis Salvage Work (AP) AP - Three weeks after the Genesis space capsule slammed into the Utah desert after its parachutes failed

to deploy, NASA scientists remain hopeful they can salvage the mission and reveal clues about the origin of the solar system."

The above document does not contain the word “Sun”. The document talks about a Space Capsule and the origin of the solar system. With only one word and no context, the model linked "Sun" to its meaning as a star. However, System C correctly ranked the relevant documents 227 and 240 first and second, above document 231. The relevant documents were about companies named Sun: Sun Microsystems (227, 240) and Sun Capital (199). Document 199 was not in System C’s top 3. This is an example for when the System chooses the wrong meaning.

2.3.5 Which types of queries are better handled by each approach?

Approach 1 - System A

System A follows the approach of word matching. It handles exact keyword queries well when the words are specific and have only one meaning. For Query 1 ("oil prices") and Query 2 ("Firefox"), their words directly appear in the relevant documents. Hence all of System A’s top 3 documents were relevant (3/3). However, System A has its limitations when it comes to other types of queries. System A does not handle Synonym Queries well. For Query 5, “joblessness” did not match "unemployment". For Query 6, System A, matched only "attacks", returning articles about heart attacks (Document 209) and car bombs (Document 18). It also does not handle word order sensitive queries as Queries 7 and 8, received identical results and scores. This is because System A ignores word order. Based on our results, System A works best with exact keyword queries whose words are specific and unambiguous.

Approach 2 - System B

System B retrieved the documents by matching pairs of neighbouring words (bigrams) as well as single words (unigrams), which makes it sensitive to word order. *This was the only system whose scores changed clearly between Query 7 and Query 8. For Query 7 ("Braves beat Astros"), System B gave document 137 a score of 0.0746 and ranked it first.* For Query 8 ("Astros beat Braves"), the same document received only 0.0435 and fell to second place. This was because Document 137 contains the word pairs, “braves beat” and “beat astros” which match only Query 7. System A by contrast gave both these queries identical scores. Our results show that Word Order Sensitive Queries are better handled by System B.

Approach 3 - System C

System C converts the document and queries into embeddings. Each of the embeddings is a vector of 384 numbers representing its meaning, using the pretrained model all-MiniLM-L6-v2. It ranks the documents by Cosine Similarity. This approach compares meaning instead of exact words. As a result, it handles Synonym Queries and Paraphrased Queries better than any of the other systems. For Query 6 ("computer virus attacks"), no document contains the word "virus", yet System C retrieved three relevant documents about spyware, malware and a browser security hole (Documents 225, 230 and 287).